NON-CALCULATOR – 30 MINUTES

SAMPLE ASSESSMENT MATERIAL

Do not open this paper until you are told to do so by the invigilator.

Overall assessment marks available: **60**
Overall assessment time limit: **2 HOURS**

There are **TWO** Sections to this assessment:

- **Section A** includes Task 1. You **must not** use a calculator for this section.

  Total marks available: 15. Time limit: 30 minutes
- **Section B** includes Task 2, 3 and 4. You can use a non-scientific calculator for this section

  Total marks available: 45. Time limit: 1 hour and 30 minutes

For Section A you need:

- This question and answer paper
- A pen with black or blue ink
- A pencil
- A ruler

INTERNET ACCESS IS NOT PERMITTED AND YOU MUST NOT USE A CALCULATOR

The invigilator will stop the assessment after 30 minutes. You must hand in this question and answer paper at this point.

The invigilator will then hand out **Section B** and a non-scientific calculator. You will then have a further 1 hour and 30 minutes to complete **Section B**.

Instructions

1. Please sign and date below to confirm that your details are correct and that you have understood the instructions.
2. Read each task and question carefully.
3. Remember to show all your workin
4. The number of marks available for rks to guide you on how long to spend on ea
5. Answer **all** questions using the spa
6. If you have time, check your work f this question and answer paper, you will n
7. If you use extra paper, write your n are answering on it and securely attach it

Learner name:	Joe Bloggs
Learner number:	10000000
Centre number:	10000
Signature:	
Today's date:	

OFFICIAL MFS MODEL SOLUTION

60 / 60 Marks (100% Pass)

· Assessment: Sample Assessment Materials (SAM)
· Complete step-by-step worked models
· Official Mark Scheme Allocations [M1 / A1]
· Exam Board: Open Awards (Ofqual Regulated)

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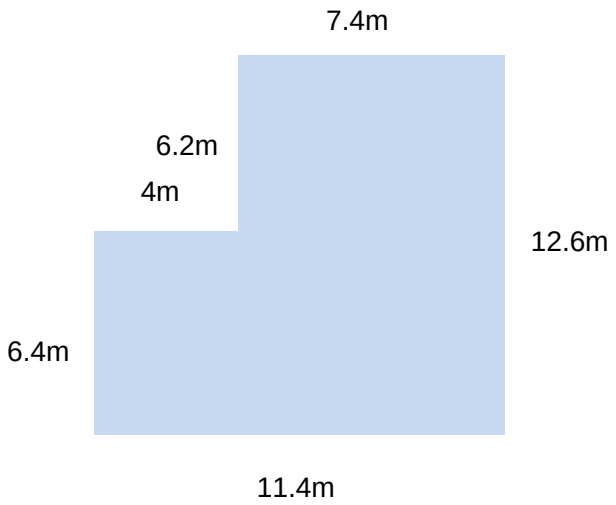
Section A

Task 1 (15 marks)

Question 1

Diagram not drawn to scale



Calculate the perimeter of the shape. (2 marks)

Show your calculations and/or workings out here:

Step 1: Sum all outer side lengths to find perimeter:

Perimeter = 7.4 + 12.6 + 11.4 + 6.2 + 4.0 + 6.4 [M1]

= 20.0 + 17.6 + 10.4 = 48.0 m [A1]

48 m

Write your answer in this box.

Question 2

Calculate 17^2. (1 mark)

Could use a grid / long multiplication:

$17^2 = 17 \times 17 = 17 \times 10 + 17 \times 7 = 170 + 119 = 289$

Show your calculations and/or workings out here: [1 mark]

289 ✓

Write your answer in this box

Question 3

There is a ratio of 6:1 red marbles to white marbles.

There are 42 red marbles in total. How many white marbles are there? (1 mark)

Compare decimals by padding to 3 decimal places:

$0.045 < 0.405 < 0.450 < 0.540$

Show your calculations and/or workings out here: [1 mark]

0.045, 0.405, 0.45, 0.54 ✓

Write your answer in this box.

Question 4

Calculate 14.73 + 18.49 (1 mark)

Show your calculations and/or workings out here:

Add numbers by aligning decimal points:

14.73 + 18.49 = 33.22

[1 mark]

33.22

Write your answer in this box.

Question 5

Calculate 46.8 ÷ 100 (1 mark)

Show your calculations and/or workings out here:

Divide by denominator 3 to find 1/3 of £450:

£450 ÷ 3 = £150

[1 mark]

£150

Write your answer in this box.

Question 6

a) Janice wants to put border strips around **three** sides of her square lawn. The length of one side of the lawn is 14 metres.

Find 20% discount on £360: 10% = £36 --> 20% = £36 × 2 = £72

£360 × 0.20 = £72.00 [1 mark]

Border strip

£72

To calculate the number of strips she needs, Janice follows this rule:

Number of border strips	=	Total length in metres	→	÷ 3	→	X 2

Each border strip costs £9.89.

Show an **estimate** of the cost of the required number of border strips. (4 marks)

Show your calculations and/or workings out here:

Subtract 20% discount from original price:

£360 - £72 = £288.00 [1 mark]

£288

Write your answer in this box.

b) Janice wants to paint the fence panels in her garden. Each fence panel needs approximately 300 ml of paint. She buys 2 tins of paint. Each tin contains 1.5 litres of paint.

How many fence panels can she paint before she runs out? (2 marks)

Show your calculations and/or workings out here:

Step 1: Calculate total area of 8 fence panels (1.8m × 1.2m each):

1 panel = 1.8m × 1.2m = 2.16 m² [M1]

Total area = 8 × 2.16 m² = 17.28 m² [M1]

Step 2: Each tin covers 10 m². Find tins needed:

17.28 ÷ 10 = 1.728 tins --> Must round up to 2 whole tins [M1 A1]

Write your answer in this box.

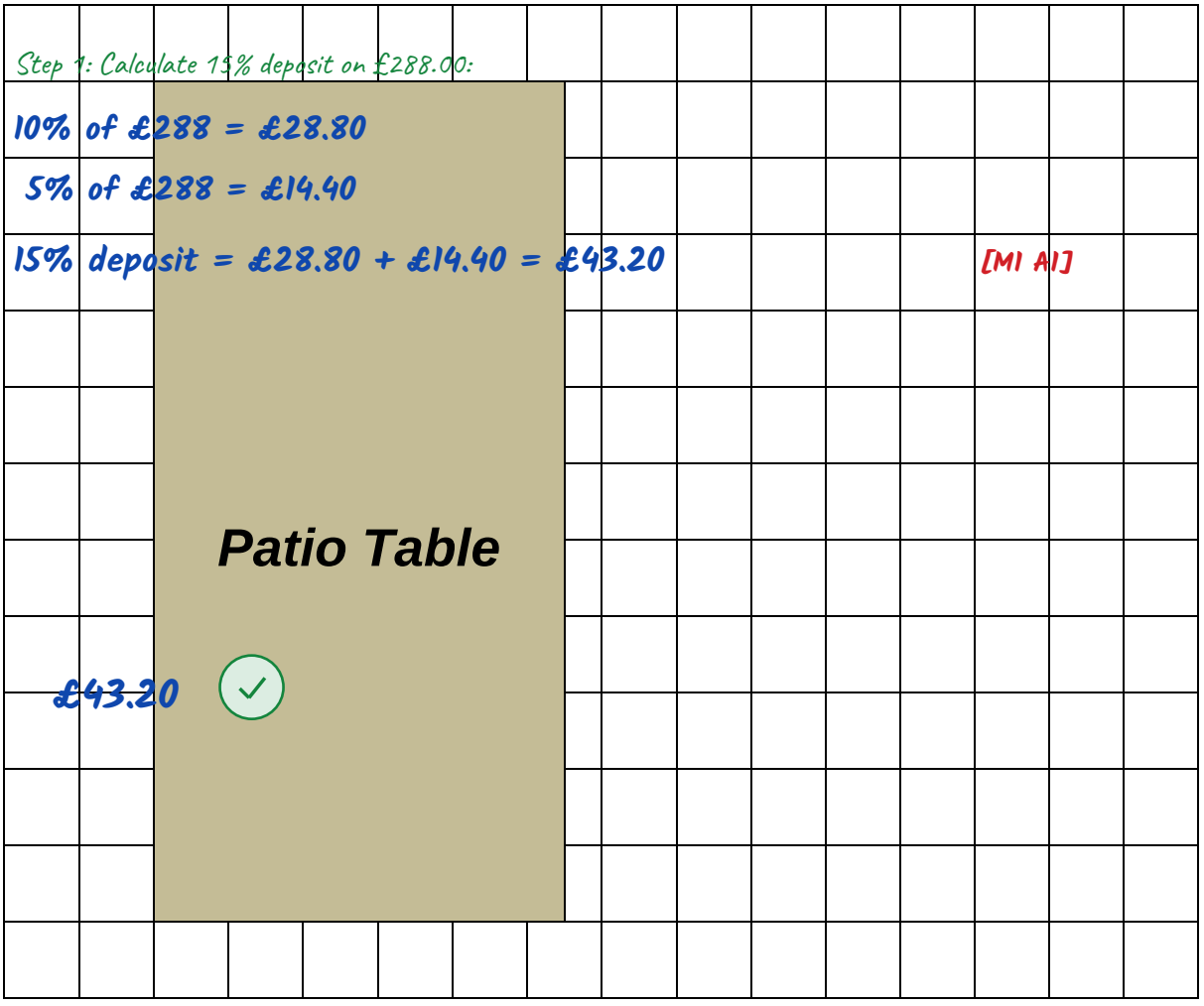
2 tins of paint ✓

c) Janice wants to buy a table to put in her patio area. She draws a diagram of her patio to show the size of table she wants.

What are the actual measurements of the table Janice wants? (2 marks)

Scale

1 box = 20cm x 20 cm



Show your calculations and/or workings out here:

Approximation check: round £360 to £350 and find 20%:

20% of £350 = 0.2 × 350 = £70 [1 mark]

Estimated price = £350 − £70 = £280 (close to £288)

£350 − £70 = £280 ✓

Write your answers in this box.

Question 7

Janice wants to buy new carpet for her bedroom. The bedroom is rectangular and has a floor area of $30m^2$.

Complete the table below to show the dimensions of her bedroom. (1 mark)

Bedroom	
Length	
Width	4m

Step 1: Calculate area of rectangular path (Length × Width):

Area = 12m × 1.5m = 18 m² [M1]

Show your calculations and/or workings out here:

Step 2: Each slab has area 0.25 m². Divide total area by slab area:

18 ÷ 0.25 = 18 × 4 = 72 slabs [M1 A1]

72 paving slabs ✓

[End of Section A]

LEVEL 1 FUNCTIONAL SKILLS QUALIFICATION IN MATHEMATICS

SECTION B - QUESTION AND ANSWER PAPER (RFSML1SAM01)

CALCULATOR – 1 HOUR AND 30 MINUTES

SAMPLE ASSESSMENT MATERIAL

Do not open this paper until you are told to do so by the invigilator.

Overall assessment marks available: **60**
Overall assessment time limit: **2 HOURS**

There are **TWO** Sections to this assessment:

- **Section A** – please ensure you have handed in Section A before beginning Section B
- **Section B** includes Task 2, 3 and 4. You can use a non-scientific calculator for this section.

Total marks available: 45. Time limit: 1 hour and 30 minutes.

For Section B you need:

- This question and answer paper
- A pen with black or blue ink
- A pencil
- A ruler
- A non-scientific calculator

INTERNET ACCESS IS NOT PERMITTED

You now have a further 1 hour and 30 minutes to complete **Section B**.

Instructions

1. Please sign and date below to confirm that your details are correct and that you have understood the instructions.
2. Read each task and question carefully.
3. Remember to show all your workings out clearly.
4. The number of marks available for each question is shown in brackets. Use these marks to guide you on how long to spend on each question.
5. Answer **all** questions using the space provided on this question and answer paper.
6. If you have time, check your work for **Section B** at the end.
7. If you use extra paper, write your name, learner number and the question number you are answering on it, and securely attach it to this question and answer paper.
8. At the end of this section (**Section B**), hand in this question and answer paper and all notes to the invigilator.

Learner name:	Joe Bloggs
Learner number:	10000000
Centre number:	10000
Signature:	
Today's date:	

Section B

Task 2 (15 marks)

Divide whole numbers:

Question 8

456 ÷ 8 = 57

A bag contains 6 red balls, 6 yellow balls and 6 blue balls. Express as a fraction the probability of picking out a blue ball. [1 mark] (1 mark)

Show your calculations and/or workings out here:

57 ✓

Write your answer in this box.

Question 9

Probability of rolling a 4 on a standard 6-sided dice:

Convert $\frac{5}{8}$ to a percentage.

P(4) = 1 out of 6 = 1/6 [1 mark] (1 mark)

Show your calculations and/or workings out here:

1/6 ✓

Write your answer in this box.

Question 10

Calculate 2.5 ÷ 0.375.

Give your answer to two decimal places. (1 mark)

Show your calculations and/or workings out here:

Step 1: Total time taken = (6 loaves × 7 mins) + 45m proving + 10m cooling:

Time = 42 + 45 + 10 = 97 minutes (1 hour 37 minutes) [M1 A1]

Step 2: Count back 1 hour 37 minutes from 6:15 am:

6:15 am - 1 hour = 5:15 am --> 5:15 am - 37 mins = 4:38 am [M1 A1]

Write your answer in this box.

4:38 am ✓

Question 11

a) Jared owns a bakery. He receives an order for party cakes to serve 72 people. He has this list of ingredients:

Party Cake Ingredients
(Serves 12 people)
400g flour
400g butter
375g sugar
6 eggs

Step 1: Ratio of flour to water is 5 : 2 for 500g of flour:

1 part = 500g ÷ 5 = 100g

Water needed = 2 parts x 100g = 200g (or 200 ml) [M1 A1]

Jared has 2 ¾ kg of flour.

How much flour will Jared have left over if he bakes enough cakes for everyone at the party? Give the units in your answer. (4 marks)

Show your calculations and/or workings out here:

200 g (200 ml) ✓

Write your answer in this box.

b) Jared wants to bake 6 loaves of bread. A loaf of bread takes 7 minutes to prepare. Jared will bake all the loaves at the same time. The loaves will take 45 minutes to bake. He allows 10 minutes for them to cool down.

What is the latest time that Jared should start preparing his first loaf of bread so that all the loaves are cooled by 6.15am?

(3 marks)

Show your calculations and/or workings out here:

Step 1: Total flour for 6 loaves @ 500g each:

6 × 500g = 3000g = 3.0 kg [M1]

Step 2: Divide by bag size 1.5 kg:

3 kg ÷ 1.5 kg = 2 bags of flour [M1 A1]

2 bags ✓

Write your answer in this box.

c) Jared needs to buy decorations for a celebration cake. He sees this price list at his local trade store.

Item	Price
Candles	13p each
Icing	£1.40 per packet
Sugar Alphabet Letters	56p each
20% discount if you use your trade card	

Step 1: Calculate total cost of items before discount:

Jared will use his trade card to buy the decorations. He buys 20 candles, 2 packs of icing and 15 letters to make the word CONGRATULATIONS.

- 20 candles @ £0.13 = £2.60
- 2 packs icing @ £1.40 = £2.80
- 15 letters @ £0.56 = £8.40

How much more will Jared spend on the decorations using his trade card?

Total = £2.60 + £2.80 + £8.40 = £13.80 [M1] (4 marks)

Show your calculations and/or workings out here:

Step 2: Apply 20% discount: £13.80 × 0.20 = £2.76 discount

Final price = £13.80 − £2.76 = £11.04 [M1 A1]

£11.04 ✓

Write your answer in this box.

d) Use approximation to check the amount of the trade discount. (1 mark)

Approximation check: round £13.80 to £14.00 and find 20%:

Show your calculations and/or workings out here:

20% of £14 = 0.2 × 14 = £2.80 [1 mark]

£2.80 (0.2 × £14) ✓

Write your answer in this box.

Task 3 (15 marks)

Question 12

Write one hundred and ninety thousand, four hundred and ninety three in digits:

Write 'One hundred and ninety thousand, four hundred and ninety three' in numbers.

190,493 [1 mark] (1 mark)

Write your answer in this box.

190,493

Question 13

The table below shows interest rates from 5 different banks.

Bank A	Bank B	Bank C	Bank D	Bank E
4.72%	4.07%	4.726%	4.672%	4.76%

Which bank offers the highest interest rate? (1 mark)

Write your answer in this box.

Compare decimals: 4.76% > 4.726% > 4.72% > 4.672% > 4.07%:

Bank E offers 4.76%, which is the highest interest rate. [1 mark]

Bank E (4.76%)

Question 14

The table below shows the scores of 16 students in an exam.

42	33	28	19	35	41	36
38	31	29	27	44	40	28
39	14					

Tally the 16 test scores into grouped intervals:

Complete the table below to show frequency. (1 mark)

Number of marks	Frequency
0 – 9	
10 – 19	
20 – 29	
30 – 39	
40 – 49	

0 ✓

2 (19, 14)

4 (28, 29, 27, 28) [1 mark]

6 (33, 35, 36, 38, 31, 39)

4 (42, 41, 44, 40)

Question 15

a) Ryan is a member of a five a side football team.

The table below shows the goal difference for Ryan's team over the last 4 football matches.

Goal difference is the number of 'goals for' minus the number of 'goals against'.

Complete the table below to show the missing values. (3 marks)

Match	Goals For	Goals Against	Goal Difference
Match 1	2		-1
Match 2		2	0
Match 3	1	-6	-3
Match 4	1	3	
Totals			

[3 marks]

b) Ryan's team are playing a match on Saturday. The table below shows the predicted chance of rainfall on a scale of 0 to 1 from Monday to Saturday.

	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
Predicted chance of rainfall	0.40	0.50	0.35	0	0.92	0.88

Ryan thinks it is unlikely to rain on Saturday.

P(rain on Saturday) = 0.88, which is very close to 1 (high likelihood):

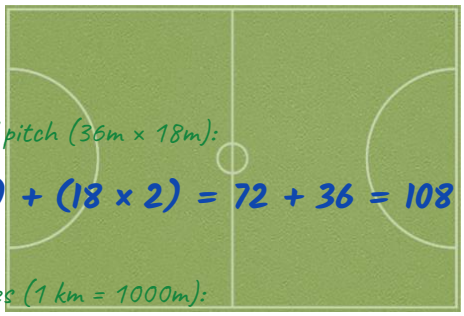
Is Ryan correct? Explain why you think this. (1 mark)

No, Ryan is incorrect because 0.88 means rain is very likely. [1 mark]

Write your answer in this box.

No, rain is likely (probability 0.88)

c) The players in Ryan's team warm up before the charity match by running at least 1km around the edge of the pitch. How many full laps of the pitch will they need to run?
(3 marks)



Step 1: Calculate perimeter of pitch (36m × 18m):

Perimeter = (36 × 2) + (18 × 2) = 72 + 36 = 108 metres [M1]

Step 2: Convert 1 km to metres (1 km = 1000m):

1000m ÷ 108m = 9.259... laps [M1]

Show your calculations and/or workings out here:

Must complete full laps: 9 laps = 972m (< 1km) --> Need 10 full laps [A1]

Write your answer in this box.

10 full laps ✓

d) Ryan's team sold 380 tickets for the charity football match.

35% of the tickets sold were for children and 20% of the tickets were sold for senior citizens. The rest of the tickets were sold at full price for adults.

Ryan thinks that 176 tickets were sold at full price for adults.

Is Ryan correct? (3 marks)

Show your calculations and/or workings out here:

Step 1: Calculate percentage of adult tickets: 100% - (35% + 20%) = 45%

Number of adult tickets = 380 × 0.45 = 171 tickets [M1 A1]

Ryan predicted 176 tickets. 171 != 176, so Ryan is NOT correct. [A1]

Write your answer in this box.

No, Ryan is incorrect (171 adult tickets sold)

e) Ryan has collected £3697.40 from the 380 ticket sales. The football club will donate £2.94 from each ticket to charity.

How much money will be left after the donation? (2 marks)

Step 1: Total donation = 380 tickets × £2.94 = £1117.20

Show your calculations and/or workings out here:

Step 2: Remainder = £3697.40 - £1117.20 = £2580.20 [M1 A1]

£2580.20

Write your answer in this box.

Task 4 (15 marks)

Question 16

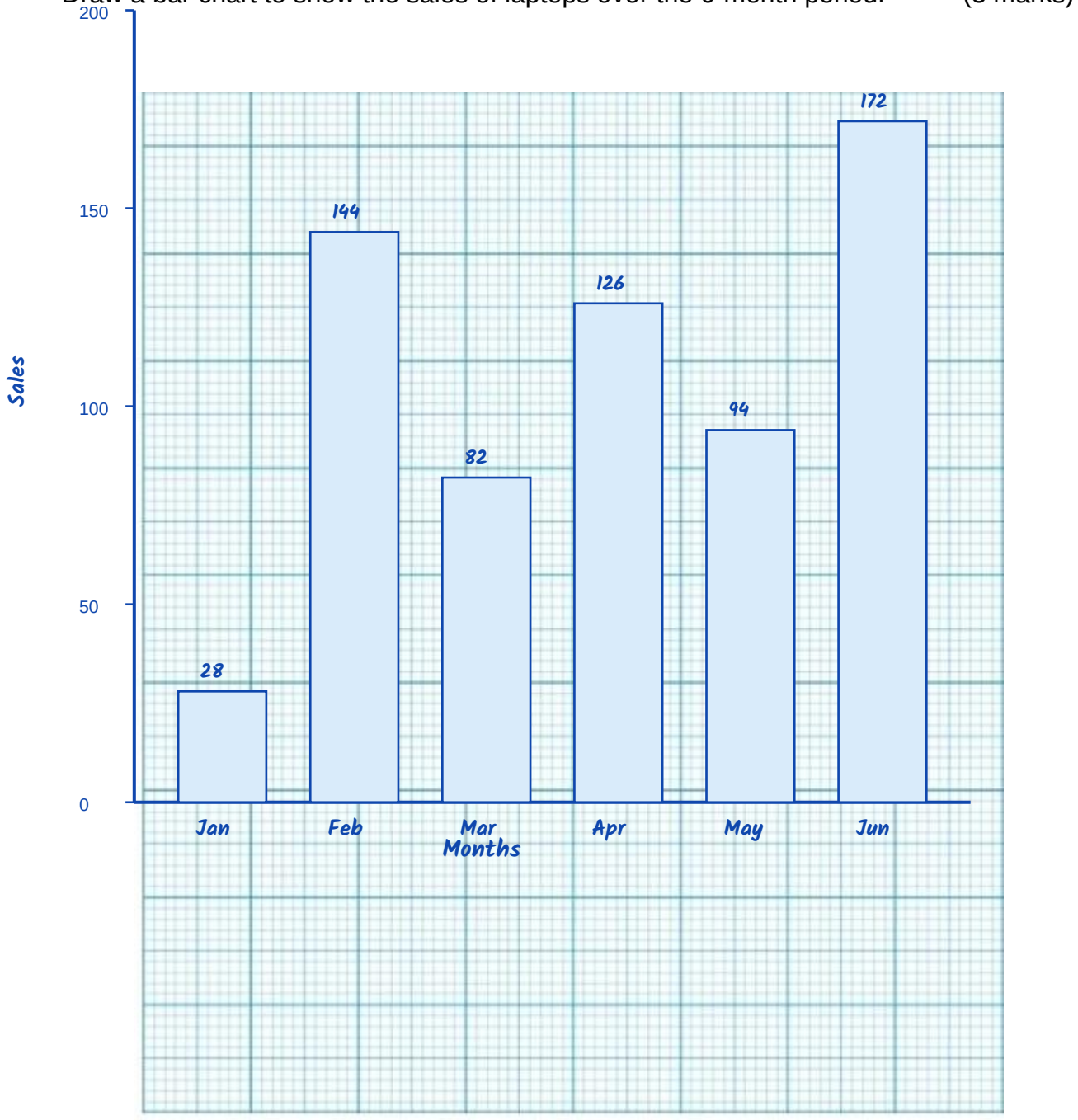
The table below shows sales of laptops over a 6-month period.

Sales of laptops						
Months	Jan	Feb	Mar	Apr	May	June
Number of laptops sold	28	144	82	126	94	172

Bar Chart: Laptop sales over 6 months (Scale: 0 to 200 in intervals of 20):

[3 marks]

Draw a bar chart to show the sales of laptops over the 6-month period. (3 marks)

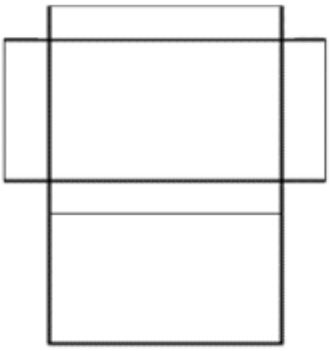


Question 17

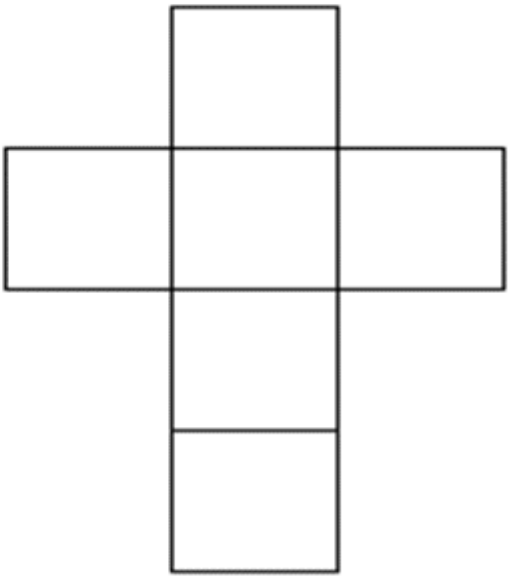
a) Davinder makes jewellery that she sells to local markets. She wants to buy some small boxes with lids, like the picture below, to put necklaces in.

She receives a selection of boxes to choose from. The net diagram of each box is shown below.

Net A
(7cm x 5cm x 1.5cm)



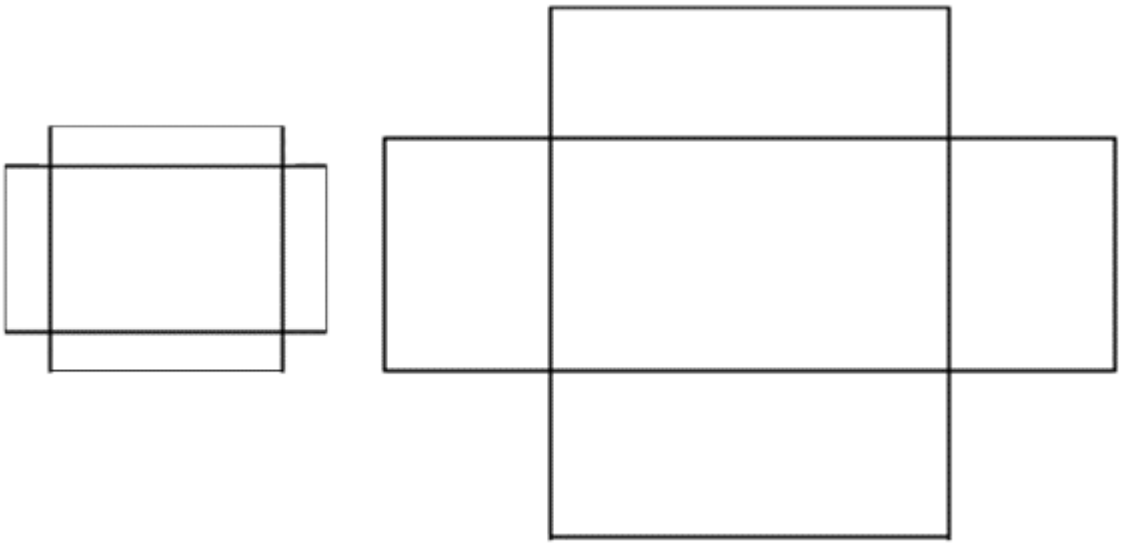
Net B
(4cm x 4cm x 4cm)



Net A has dimensions 7cm x 5cm x 1.5cm with shallow 1.5cm depth.

Net C
(7cm x 5cm x3cm)

Net D
(*10cm x 8cm x 3cm*)



Which net diagram best represents the box shown in the picture? Explain the reason for your choice of net in the box below. (2 marks)

Reason for choice of Net A:

- Net A is rectangular with very small height (1.5cm), matching shallow necklace box. [M1 A1]
- Net B forms a cube (4x4x4), Net C has no lid, Net D is too high/deep.

Net A (shallow depth 1.5cm matches picture) ✓

b) Davinder has received an order for 100 **bracelets**.

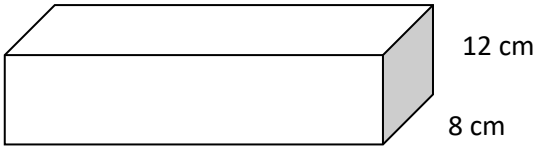
She packages each bracelet into a box 10cm by 8cm by 3cm.

Davinder will put the 100 boxes of bracelets into larger boxes for postage and delivery.

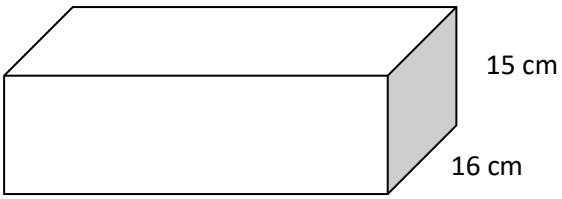
She has a choice of Box A or Box B, with measurements as shown below. She will only use one size of box.

Box A costs £0.70 each and Box B costs £1.80 each. She wants to spend as little as possible.

Box A



Box B



Should Davinder choose Box A or Box B? (4 marks)

Show your calculations and/or workings out in the box at the top of the next page.

See full calculations and comparison on top of next page -->

Show your calculations and/or workings out here:

Step 1: Calculate how many small boxes (10cm × 8cm × 3cm) fit in Box A and Box B:

- Box A (50 × 8 × 12): (50÷10) × (8÷8) × (12÷3) = 5 × 1 × 4 = 20 small boxes [M1 A1]
- Box B (50 × 15 × 16): (50÷10) × (16÷8) × (15÷3) = 5 × 2 × 5 = 50 small boxes

Step 2: Number of boxes needed for 100 bracelets and total cost:

- Box A: 100 ÷ 20 = 5 boxes needed @ £0.70 = £3.50
- Box B: 100 ÷ 50 = 2 boxes needed @ £1.80 = £3.60

[M1 A1]

£3.50 < £3.60, so Box A is cheaper.

Davinder should choose Box A (£3.50 vs £3.60) ✓

Question 18

a) Davinder wants to know what her best-selling item of jewellery is and which jewellery item sold the most consistently over the 6-month period from July to December.

She has this information:

Jewellery item	Mean of items sold per month	Range of items sold per month
Bracelets		
Necklaces	47	84
Rings	35	62
Earrings	39	71

The table below shows the sales of bracelets over the 6-month period:

Months	Jul	Aug	Sept	Oct	Nov	Dec
Number of Bracelets Sold	23	22	38	28	44	97

Which is the best-selling jewellery item over the 6-month period? (2 marks)

Show your calculations and/or workings out here:

Step 1: Calculate mean sales of bracelets over 6 months:

Total bracelets = 23 + 22 + 38 + 28 + 44 + 97 = 252

Mean bracelets = 252 ÷ 6 = 42 per month [M1]

Compare with other means: Necklaces mean = 47 (highest mean sales) [A1]

Necklaces (mean 47 items/mont ✓

Write your answer in this box.

b) Which jewellery item sold the most consistently over the 6-month period?

(2 marks)

Show your calculations and/or workings out here:

Step 1: Calculate range of bracelets sold:

Range = Maximum - Minimum = 97 - 22 = 75 [M1]

Step 2: Compare ranges to find lowest range (most consistent):

Necklaces: 84, Bracelets: 75, Earrings: 71, Rings: 62

Rings have the smallest range (62), so are the most consistent. [A1]

Write your answer in this box.

Rings (lowest range of 62) ✓

c) Davinder makes a profit of £1592 from jewellery sales. She wants to use a third of the profit to buy more stock and pay the remainder into a savings account.

How much money will Davinder pay into the savings account? (2 marks)

Show your calculations and/or workings out here:

Step 1: Davinder spends 1/3 of profit on stock and saves the rest (2/3):

Savings = £1592 × 2/3 = £3184 ÷ 3 [M1]

= £1061.333... --> £1061.33 [A1]

Write your answer in this box.

£1061.33

[End of assessment]

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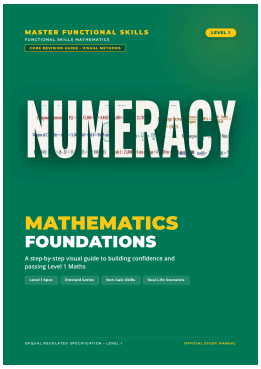

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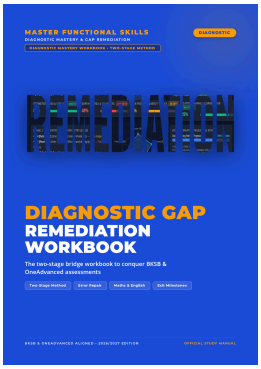

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